

Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Contrasts on the Exchange-Aligned Data Model

PENPAL Project

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This notebook compares **valence alignment** across the three experimental conditions using the new exchange-aligned structure.

- **Human-AI:** Human co-writing with an LLM
- **Human-Human:** Two humans co-writing together
- **AI-AI:** Two AI agents co-writing together

Important structural note:

- `author_1` / `author_2` are now treated as **slot 1** / **slot 2** in the cleaned exchange sequence.
- In **Human-AI**, those slots also map onto human vs AI roles.
- In **Human-Human** and **AI-AI**, the slots are positional sides of the exchange, not stable identity labels.

The analysis window is restricted to **complete exchanges** and the default aligned turn range (`analysis_turn` 1 to 9).

Primary alignment object. Each story contributes one unsigned, label-invariant dyad asymmetry magnitude:

$$A_{\text{valence}}(i) = |\text{atanh}(r_{\text{first} \rightarrow \text{second}, i}) - \text{atanh}(r_{\text{second} \rightarrow \text{first}, i})|$$

where $r_{\text{first} \rightarrow \text{second}, i}$ is the per-story Pearson correlation between same-row chronological first and second turns, and $r_{\text{second} \rightarrow \text{first}, i}$ is the correlation between the previous row's chronological second turn and the current row's chronological first turn. The main question is whether A_{valence} differs across conditions. The directional-slope LMMs are retained as supplemental views. Starter-mechanism and Human-AI speaker-type decompositions are deferred to a follow-up pass.

Primary baseline-valence object. Before the alignment analysis, we also test agent-side baseline valence asymmetry. Each story contributes one unsigned baseline gap:

$$A_{\text{baseline}}(i) = |\bar{v}_{\text{slot } 1, i} - \bar{v}_{\text{slot } 2, i}|$$

where $\bar{v}_{\text{slot } 1, i}$ and $\bar{v}_{\text{slot } 2, i}$ are the mean projected valence scores for the two sides of story i . This is a baseline-level difference, not a turn-by-turn adaptation measure.

Article-use note. The primary cross-condition claim should use the unsigned per-story A_{valence} test below. Signed directional models are diagnostic only: in Human-Human and AI-AI, chronological first / second tracks starter-side ordering rather than theoretically meaningful agent identities.

1 Load Data

```
cat("Data availability:\n")
```

```
## Data availability:
```

```
print(check_data_availability())
```

```
## # A tibble: 6 x 4
```

## file	`human-ai`	`human-human`	`ai-ai`
## <chr>	<lgl>	<lgl>	<lgl>
## 1 dyadic_sentiment_scores.parquet	TRUE	TRUE	TRUE
## 2 interaction_level_surface_metrics.parquet	TRUE	TRUE	TRUE
## 3 novelty_scores.csv	TRUE	TRUE	TRUE
## 4 semantic_exploration_binned.parquet	TRUE	TRUE	TRUE
## 5 story_embeddings_interaction_level.parquet	TRUE	TRUE	TRUE
## 6 interaction_level_stories_filtered.csv	TRUE	TRUE	TRUE

```
df_combined <- load_valence_data(c("human-ai", "human-human", "ai-ai")) %>%
  filter_complete_exchange_window() %>%
  prepare_condition_factor(reference = "human-ai") %>%
  add_chronological_valence() %>%
  filter(conversation_id != "conv_ed575a06c11d42358e3eeb7826d2f959", conversation_id != "conv_a63a08273")
```

```
df_combined %>%
  group_by(condition) %>%
  summarise(
    n_stories = n_distinct(conversation_id),
    n_turns = n(),
    min_turn = min(turn_index, na.rm = TRUE),
    max_turn = max(turn_index, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  knitr::kable()
```

```
caption = "Loaded exchange-aligned valence data by condition"
)
```

Table 1: Loaded exchange-aligned valence data by condition

condition	n_stories	n_turns	min_turn	max_turn
Human-AI	98	781	1	8
Human-Human	36	341	1	9
AI-AI	80	720	1	9

2 Primary Baseline Valence Asymmetry

This section tests whether the two sides of a story differ in their average valence baseline. The signed slot gap is only role-interpretable in Human-AI; for the cross-condition comparison, the unsigned per-story gap A_{baseline} is the main quantity.

```
df_baseline_story <- df_combined %>%
  group_by(condition, conversation_id) %>%
  summarise(
    n_exchanges = sum(!is.na(author_1_valence) & !is.na(author_2_valence)),
    slot_1_baseline = mean(author_1_valence, na.rm = TRUE),
    slot_2_baseline = mean(author_2_valence, na.rm = TRUE),
    story_baseline = mean(c(author_1_valence, author_2_valence), na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    baseline_gap = slot_1_baseline - slot_2_baseline,
    A_baseline = abs(baseline_gap)
  ) %>%
  drop_na(A_baseline, story_baseline)

df_baseline_story %>%
  group_by(condition) %>%
  summarise(
    n_stories = n(),
    mean_A_baseline = mean(A_baseline),
    sd_A_baseline = sd(A_baseline),
    median_A_baseline = median(A_baseline),
    mean_signed_gap = mean(baseline_gap),
    .groups = "drop"
  ) %>%
  knitr::kable(
    digits = 3,
    caption = "Per-story baseline valence asymmetry by condition"
  )
```

Table 2: Per-story baseline valence asymmetry by condition

condition	n_stories	mean_A_baseline	sd_A_baseline	median_A_baseline	mean_signed_gap
Human-AI	98	0.174	0.142	0.134	-0.100
Human-Human	36	0.122	0.090	0.097	-0.004

condition	n_stories	mean_A_baseline	sd_A_baseline	median_A_baseline	mean_signed_gap
AI-AI	80	0.084	0.063	0.075	-0.018

```

cat("One-way ANOVA: A_baseline ~ condition\n")

## One-way ANOVA: A_baseline ~ condition
print(summary(aov(A_baseline ~ condition, data = df_baseline_story)))

##           Df Sum Sq Mean Sq F value    Pr(>F)
## condition     2  0.3586  0.17931    14.91 8.79e-07 ***
## Residuals   211  2.5379  0.01203
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

cat("\nKruskal-Wallis (robust): A_baseline ~ condition\n")

##
## Kruskal-Wallis (robust): A_baseline ~ condition
print(kruskal.test(A_baseline ~ condition, data = df_baseline_story))

##
## Kruskal-Wallis rank sum test
##
## data:  A_baseline by condition
## Kruskal-Wallis chi-squared = 17.876, df = 2, p-value = 0.0001313

cat("\nPairwise Wilcoxon (BH-adjusted):\n")

##
## Pairwise Wilcoxon (BH-adjusted):
print(pairwise.wilcox.test(
  df_baseline_story$A_baseline,
  df_baseline_story$condition,
  p.adjust.method = "BH"
))

##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  df_baseline_story$A_baseline and df_baseline_story$condition
##
##           Human-AI Human-Human
## Human-Human  0.123      -
## AI-AI        9.9e-05  0.079
##
## P value adjustment method: BH

p_baseline_asym <- ggplot(df_baseline_story, aes(x = condition, y = A_baseline, fill = condition)) +
  geom_violin(alpha = 0.55, trim = FALSE) +
  geom_boxplot(width = 0.18, alpha = 0.85, outlier.alpha = 0.15) +
  geom_jitter(width = 0.08, alpha = 0.35, size = 1.4) +
  scale_fill_manual(values = condition_colors) +
  labs(
    title = "Per-story baseline valence asymmetry by condition",

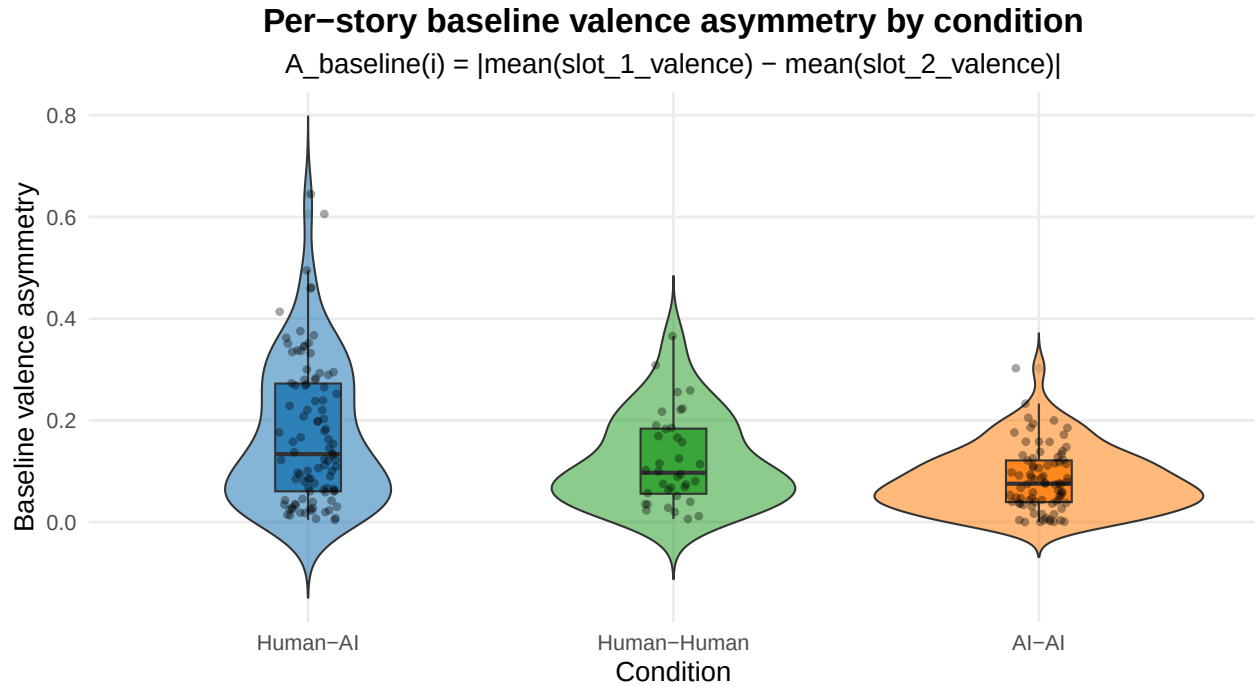
```

```

  subtitle = "A_baseline(i) = |mean(slot_1_valence) - mean(slot_2_valence)|",
  x = "Condition",
  y = "Baseline valence asymmetry"
) +
  theme(legend.position = "none")

print(p_baseline_asym)

```



```

ggsave(
  file.path(GRAPHS_DIR, "valence_baseline_asymmetry_per_story.png"),
  p_baseline_asym,
  width = 9,
  height = 5,
  dpi = 300
)

```

```

bootstrap_condition_means(df_baseline_story, "A_baseline", R = 10000) %>%
  knitr::kable(
    digits = 3,
    caption = "Bootstrap 95% CI on per-condition mean of A_baseline (R = 10,000)"
  )

```

Table 3: Bootstrap 95% CI on per-condition mean of $A_baseline$ (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	98	0.174	0.147	0.202
Human-Human	36	0.122	0.094	0.152
AI-AI	80	0.084	0.071	0.097

```
effect_sizes_pairwise(df_baseline_story, "A_baseline") %>%
  knitr::kable(
    digits = 3,
    caption = "Pairwise Cliff's delta on A_baseline"
  )
```

Table 4: Pairwise Cliff's delta on A_baseline

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	98	36	0.175	small
Human-AI vs AI-AI	98	80	0.362	medium
Human-Human vs AI-AI	36	80	0.226	small

```
planned_contrast_HA_vs_same(df_baseline_story, "A_baseline") %>%
  knitr::kable(
    digits = 3,
    caption = "Planned contrast: A_baseline in HA vs the average of HH and AA"
  )
```

Table 5: Planned contrast: A_baseline in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.071	0.017	4.291	142.263	0	0.038	0.104

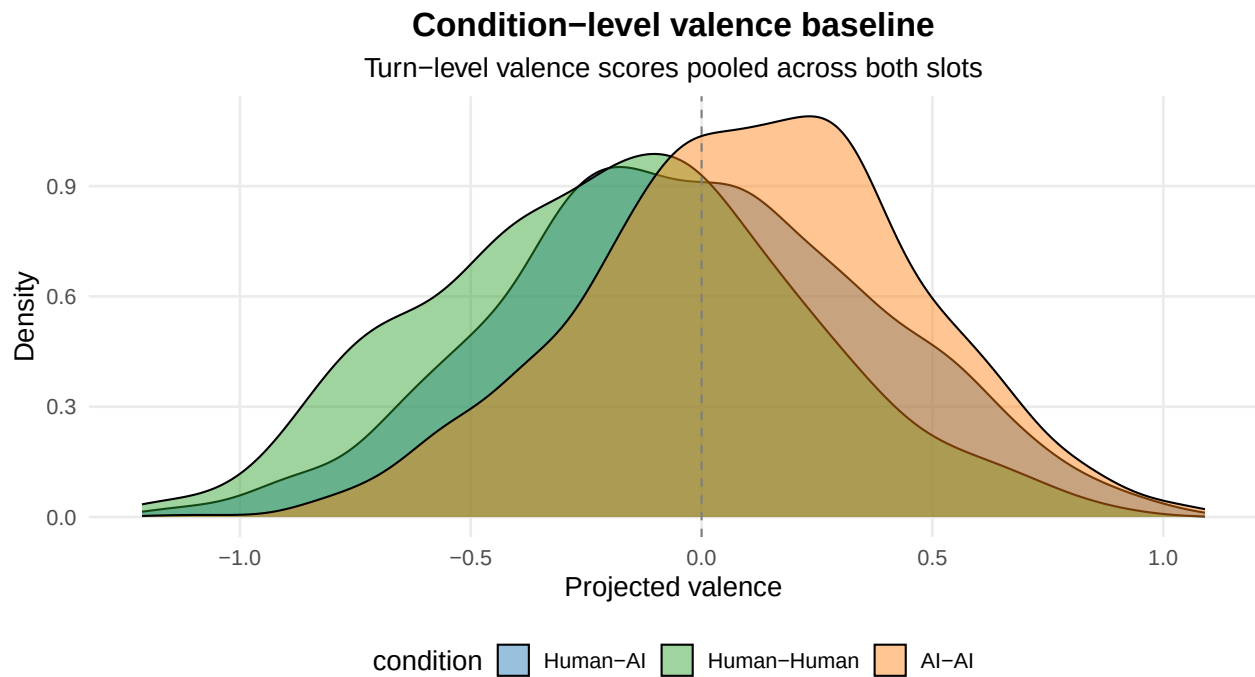
3 Condition Baseline Valence

This plot shows overall valence baseline by condition, pooling turn-level valence scores across both slots. It is not an asymmetry measure; it asks whether the conditions differ in overall emotional tone.

```
df_condition_valence <- df_combined %>%
  select(condition, conversation_id, turn_index, author_1_valence, author_2_valence) %>%
  pivot_longer(
    cols = c(author_1_valence, author_2_valence),
    names_to = "slot",
    values_to = "valence"
  ) %>%
  drop_na(valence)

p_condition_baseline <- ggplot(df_condition_valence, aes(x = valence, fill = condition)) +
  geom_density(alpha = 0.45) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  scale_fill_manual(values = condition_colors) +
  labs(
    title = "Condition-level valence baseline",
    subtitle = "Turn-level valence scores pooled across both slots",
    x = "Projected valence",
    y = "Density"
  ) +
  theme(legend.position = "bottom")
```

```
print(p_condition_baseline)
```



```
ggsave(
  file.path(GRAPHS_DIR, "valence_condition_baseline_density.png"),
  p_condition_baseline,
  width = 9,
  height = 5,
  dpi = 300
)
```

4 Preprocessing

```
df_analysis <- df_combined %>%
  group_by(conversation_id, condition) %>%
  arrange(turn_index, .by_group = TRUE) %>%
  mutate(
    second_valence_lag1 = lag(second_valence)
  ) %>%
  ungroup() %>%
  filter(turn_index > 1) %>%
  drop_na(
    first_valence, second_valence, second_valence_lag1
  )

df_alignment_long <- bind_rows(
  df_analysis %>%
    transmute(
      conversation_id,
      condition,
      turn_index,
```

```

    alignment_direction = "first_to_second",
    predictor_valence = first_valence,
    response_valence = second_valence
  ),
  df_analysis %>%
    transmute(
      conversation_id,
      condition,
      turn_index,
      alignment_direction = "second_to_first",
      predictor_valence = second_valence_lag1,
      response_valence = first_valence
    )
) %>%
  mutate(
    alignment_direction = factor(
      alignment_direction,
      levels = c("first_to_second", "second_to_first"),
      labels = c("First → Second", "Second(t-1) → First")
    )
  )

df_std <- df_alignment_long %>%
  group_by(conversation_id, alignment_direction) %>%
  mutate(
    predictor_z = as.numeric(scale(predictor_valence)),
    response_z = as.numeric(scale(response_valence))
  ) %>%
  ungroup() %>%
  drop_na(predictor_z, response_z)

contrasts(df_std$alignment_direction) <- contr.sum(2)

cat("Analysis rows:", nrow(df_std), "\n")

## Analysis rows: 3222

cat("Stories:", n_distinct(df_std$conversation_id), "\n")

## Stories: 214

```

5 Descriptive Checks

```

df_analysis %>%
  group_by(condition) %>%
  summarise(
    n_stories = n_distinct(conversation_id),
    n_turns = n(),
    slot_1_mean = mean(author_1_valence, na.rm = TRUE),
    slot_2_mean = mean(author_2_valence, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  knitr::kable(

```



```

digits = 3,
caption = "Mean projected valence by condition"
)

```

Table 6: Mean projected valence by condition

condition	n_stories	n_turns	slot_1_mean	slot_2_mean
Human-AI	98	683	-0.088	0.023
Human-Human	36	288	-0.209	-0.203
AI-AI	80	640	0.093	0.112

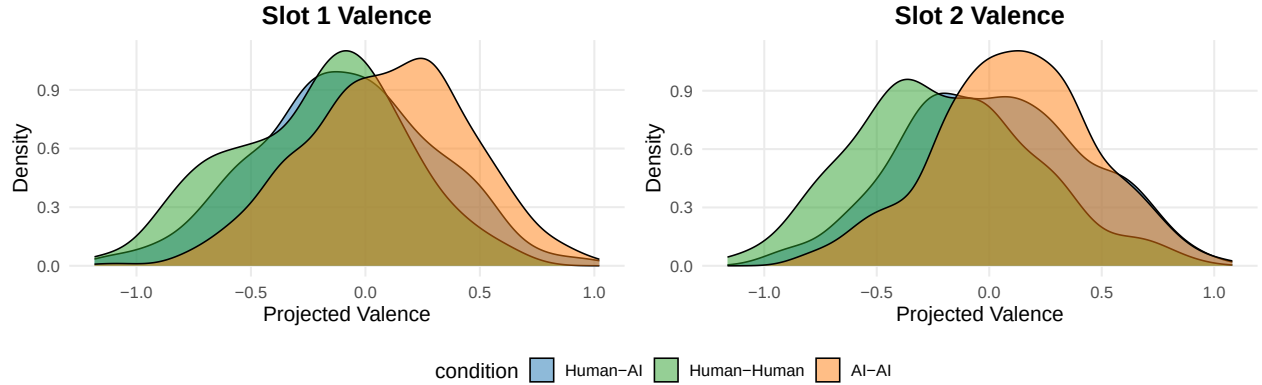
```

p_slot1 <- df_analysis %>%
  ggplot(aes(x = author_1_valence, fill = condition)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = condition_colors) +
  labs(title = "Slot 1 Valence", x = "Projected Valence", y = "Density") +
  theme(legend.position = "none")

p_slot2 <- df_analysis %>%
  ggplot(aes(x = author_2_valence, fill = condition)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = condition_colors) +
  labs(title = "Slot 2 Valence", x = "Projected Valence", y = "Density")

slot_combined <- p_slot1 + p_slot2 + plot_layout(guides = "collect")
print(slot_combined)

```



```

ggsave(
  file.path(GRAPHS_DIR, "valence_slot_distributions.png"),
  slot_combined,
  width = 12,
  height = 4,
  dpi = 300
)

```

6 Directional Valence Alignment by Condition

The figure below visualizes responder valence as a function of partner valence under the two exchange-aligned pairing rules. In Human-AI, the directions map onto Human $\rightarrow$ AI and AI $\rightarrow$ Human response patterns.

In Human-Human and AI-AI, they should be interpreted as chronological exchange positions rather than stable agent identities.

```
df_alignment_long %>%
  group_by(condition, alignment_direction) %>%
  summarise(
    n_pairs = sum(!is.na(predictor_valence) & !is.na(response_valence)),
    slope = if (n_pairs >= 3) {
      coef(lm(response_valence ~ predictor_valence))[2]
    } else {
      NA_real_
    },
    intercept = if (n_pairs >= 3) {
      coef(lm(response_valence ~ predictor_valence))[1]
    } else {
      NA_real_
    },
    .groups = "drop"
  ) %>%
  knitr::kable(
    digits = 3,
    caption = "Pooled descriptive slopes for responder valence predicted by partner valence"
  )
```

Table 7: Pooled descriptive slopes for responder valence predicted by partner valence

condition	alignment_direction	n_pairs	slope	intercept
Human-AI	First → Second	683	0.423	0.060
Human-AI	Second(t-1) → First	683	0.333	-0.094
Human-Human	First → Second	288	0.382	-0.099
Human-Human	Second(t-1) → First	288	0.374	-0.154
AI-AI	First → Second	640	0.543	0.046
AI-AI	Second(t-1) → First	640	0.513	0.061

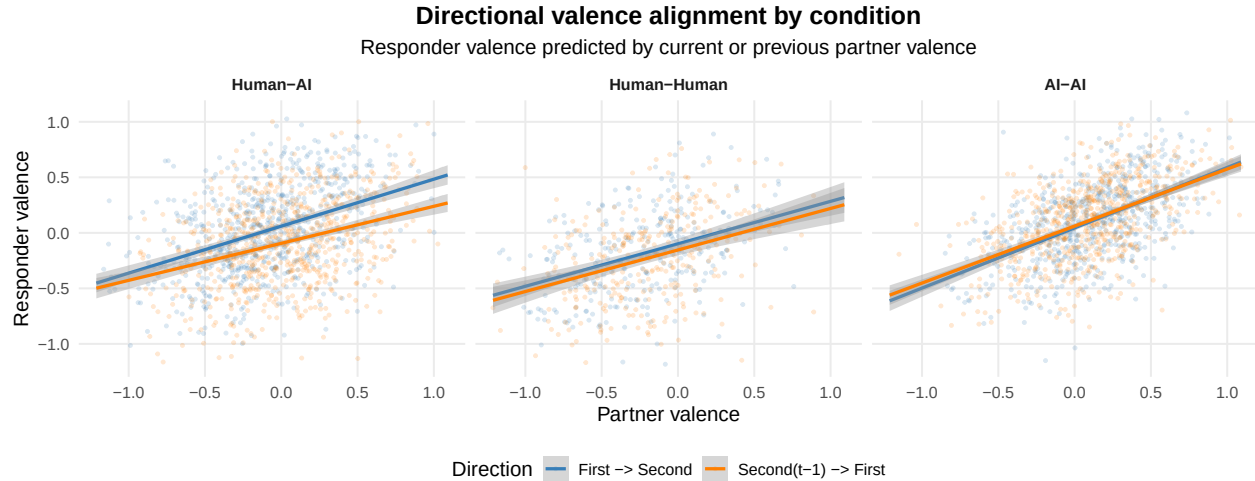
```
p_directional_alignment_comparison <- df_alignment_long %>%
  drop_na(predictor_valence, response_valence) %>%
  ggplot(aes(
    x = predictor_valence,
    y = response_valence,
    color = alignment_direction
  )) +
  geom_point(alpha = 0.18, size = 0.9) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.05, fullrange = TRUE) +
  facet_wrap(~ condition, nrow = 1) +
  scale_color_manual(
    values = c("First → Second" = "#377EB8", "Second(t-1) → First" = "#FF7F00"),
    name = "Direction"
  ) +
  labs(
    title = "Directional valence alignment by condition",
    subtitle = "Responder valence predicted by current or previous partner valence",
    x = "Partner valence",
```

```

  y = "Responder valence"
) +
  theme(legend.position = "bottom")

print(p_directional_alignment_comparison)

```



```

ggsave(
  file.path(GRAPHS_DIR, "directional_valence_alignment_comparison_three_panel.png"),
  p_directional_alignment_comparison,
  width = 12,
  height = 4.8,
  dpi = 300
)

```

7 Primary: Per-Story Asymmetry Magnitude

The three-panel figure suggests that the two directional slopes may differ most strongly in Human-AI. The mixed model below tests that visual pattern directly by estimating whether the predictor-valence slope differs by direction and whether that directional slope gap changes by condition. This is a pooled turn-level model with story-level random intercepts; the per-story A_{valence} analysis that follows is the story-level robustness layer.

```

model_direction_condition <- lmer(
  response_z ~ predictor_z * alignment_direction * condition + (1 | conversation_id),
  data = df_std
)

```

```

cat("=== Direction x condition slope-interaction model ===\n")

```

```

## === Direction x condition slope-interaction model ===

```

```

print(summary(model_direction_condition)$coefficients)

```

	Estimate	Std. Error
## (Intercept)	-4.671384e-18	0.02482269
## predictor_z	1.354038e-01	0.02682139
## alignment_direction1	5.926804e-18	0.02482269
## conditionHuman-Human	-1.157781e-17	0.04557873
## conditionAI-AI	2.478700e-17	0.03568936

```

## predictor_z:alignment_direction1          7.228943e-02 0.02682139
## predictor_z:conditionHuman-Human          9.564638e-02 0.04888141
## predictor_z:conditionAI-AI                4.363879e-02 0.03835218
## alignment_direction1:conditionHuman-Human -2.291123e-17 0.04557873
## alignment_direction1:conditionAI-AI       -3.121924e-17 0.03568936
## predictor_z:alignment_direction1:conditionHuman-Human -7.964513e-02 0.04888141
## predictor_z:alignment_direction1:conditionAI-AI -3.223223e-02 0.03835218
##                                           df      t value
## (Intercept)                          3210 -1.881901e-16
## predictor_z                          3210  5.048352e+00
## alignment_direction1                  3210  2.387655e-16
## conditionHuman-Human                  3210 -2.540178e-16
## conditionAI-AI                        3210  6.945205e-16
## predictor_z:alignment_direction1      3210  2.695215e+00
## predictor_z:conditionHuman-Human      3210  1.956702e+00
## predictor_z:conditionAI-AI            3210  1.137844e+00
## alignment_direction1:conditionHuman-Human 3210 -5.026737e-16
## alignment_direction1:conditionAI-AI      3210 -8.747490e-16
## predictor_z:alignment_direction1:conditionHuman-Human 3210 -1.629354e+00
## predictor_z:alignment_direction1:conditionAI-AI 3210 -8.404277e-01
##                                           Pr(>|t|)
## (Intercept)                          1.000000e+00
## predictor_z                          4.704660e-07
## alignment_direction1                  1.000000e+00
## conditionHuman-Human                  1.000000e+00
## conditionAI-AI                        1.000000e+00
## predictor_z:alignment_direction1      7.070980e-03
## predictor_z:conditionHuman-Human      5.046905e-02
## predictor_z:conditionAI-AI            2.552706e-01
## alignment_direction1:conditionHuman-Human 1.000000e+00
## alignment_direction1:conditionAI-AI      1.000000e+00
## predictor_z:alignment_direction1:conditionHuman-Human 1.033362e-01
## predictor_z:alignment_direction1:conditionAI-AI 4.007312e-01

directional_slopes <- emtrends(
  model_direction_condition,
  specs = c("condition", "alignment_direction"),
  var = "predictor_z"
)

directional_slopes %>%
  as.data.frame() %>%
  knitr::kable(
    digits = 3,
    caption = "Estimated predictor-valence slopes by condition and direction"
  )

```

Table 8: Estimated predictor-valence slopes by condition and direction

condition	alignment_direction	predictor_z.trend	SE	df	asymp.LCL	asymp.UCL
Human-AI	First → Second	0.208	0.038	Inf	0.133	0.282
Human-Human	First → Second	0.224	0.058	Inf	0.110	0.337

condition	alignment_direction	predictor_z.trend	SE	df	asympt.LCL	asympt.UCL
AI-AI	First $\rightarrow$ Second	0.219	0.039	Inf	0.143	0.295
Human-AI	Second(t-1) $\rightarrow$ First	0.063	0.038	Inf	-0.011	0.137
Human-Human	Second(t-1) $\rightarrow$ First	0.238	0.058	Inf	0.125	0.352
AI-AI	Second(t-1) $\rightarrow$ First	0.139	0.039	Inf	0.063	0.215

```

direction_gap_by_condition <- contrast(
  directional_slopes,
  method = list("First_to_second_minus_second_to_first" = c(1, -1)),
  by = "condition"
)

direction_gap_by_condition %>%
  as.data.frame() %>%
  knitr::kable(
    digits = 3,
    caption = "Within-condition directional slope gap"
  )

```

Table 9: Within-condition directional slope gap

contrast	condition	estimate	SE	df	z.ratio	p.value
First_to_second_minus_second_to_first	Human-AI	0.145	0.054	Inf	2.695	0.007
First_to_second_minus_second_to_first	Human-Human	-0.015	0.082	Inf	-0.180	0.857
First_to_second_minus_second_to_first	AI-AI	0.080	0.055	Inf	1.461	0.144

```

direction_gap_slopes <- contrast(
  directional_slopes,
  method = list("First_to_second_minus_second_to_first" = c(1, -1)),
  by = "condition"
)

contrast(
  direction_gap_slopes,
  method = list(
    "Human-AI vs Human-Human" = c(1, -1, 0),
    "Human-AI vs AI-AI" = c(1, 0, -1),
    "Human-AI vs same-type average" = c(1, -0.5, -0.5)
  ),
  by = NULL,
  adjust = "none"
) %>%
  as.data.frame() %>%
  knitr::kable(
    digits = 3,
    caption = "Cross-condition contrasts of the directional slope gap"
  )

```

Table 10: Cross-condition contrasts of the directional slope gap

contrast	estimate	SE	df	z.ratio	p.value
Human-AI vs Human-Human	0.159	0.098	Inf	1.629	0.103
Human-AI vs AI-AI	0.064	0.077	Inf	0.840	0.401
Human-AI vs same-type average	0.112	0.073	Inf	1.537	0.124

```

# Per-story Pearson correlations under both pairing rules, on non-standardized
# predictor/response pairs (r is scale-invariant).
min_pairs <- 3

df_r12 <- df_alignment_long %>%
  filter(alignment_direction == "First \u2192 Second") %>%
  group_by(conversation_id, condition) %>%
  summarise(
    n_pairs = sum(!is.na(predictor_valence) & !is.na(response_valence)),
    r = if (n_pairs >= min_pairs) {
      suppressWarnings(cor(predictor_valence, response_valence, use = "complete.obs"))
    } else {
      NA_real_
    },
    .groups = "drop"
  ) %>%
  transmute(conversation_id, condition, n_pairs_12 = n_pairs, r_12 = r)

df_r21 <- df_alignment_long %>%
  filter(alignment_direction == "Second(t-1) \u2192 First") %>%
  group_by(conversation_id, condition) %>%
  summarise(
    n_pairs = sum(!is.na(predictor_valence) & !is.na(response_valence)),
    r = if (n_pairs >= min_pairs) {
      suppressWarnings(cor(predictor_valence, response_valence, use = "complete.obs"))
    } else {
      NA_real_
    },
    .groups = "drop"
  ) %>%
  transmute(conversation_id, condition, n_pairs_21 = n_pairs, r_21 = r)

df_asym <- df_r12 %>%
  inner_join(df_r21, by = c("conversation_id", "condition")) %>%
  mutate(
    # Clamp |r|<1 before atanh to avoid infinities on degenerate stories
    r_12_c = pmin(pmax(r_12, -0.999999), 0.999999),
    r_21_c = pmin(pmax(r_21, -0.999999), 0.999999),
    z_12 = atanh(r_12_c),
    z_21 = atanh(r_21_c),
    signed_z_diff = z_12 - z_21,
    A_valence = abs(signed_z_diff)
  ) %>%
  filter(!is.na(A_valence))

cat("Stories with both directional correlations available:\n")

```

```
## Stories with both directional correlations available:
df_asym %>%
  count(condition) %>%
  knitr::kable(caption = "Per-story A_valence sample sizes by condition")
```

Table 11: Per-story A_valence sample sizes by condition

condition	n
Human-AI	98
Human-Human	36
AI-AI	80

```
# Sanity checks
stopifnot(all(df_asym$A_valence >= 0))

df_asym %>%
  group_by(condition) %>%
  summarise(
    n = n(),
    mean_A = mean(A_valence),
    sd_A = sd(A_valence),
    median_A = median(A_valence),
    .groups = "drop"
  ) %>%
  knitr::kable(
    digits = 3,
    caption = "Per-story |Fisher-z asymmetry| by condition"
  )
```

Table 12: Per-story |Fisher-z asymmetry| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	98	0.612	0.471	0.554
Human-Human	36	0.516	0.405	0.439
AI-AI	80	0.490	0.355	0.425

```
cat("One-way ANOVA: A_valence ~ condition\n")

## One-way ANOVA: A_valence ~ condition
print(summary(aov(A_valence ~ condition, data = df_asym)))

##           Df Sum Sq Mean Sq F value Pr(>F)
## condition    2   0.71  0.3534   2.003  0.137
## Residuals  211  37.22  0.1764

cat("\nKruskal-Wallis (robust): A_valence ~ condition\n")

##
## Kruskal-Wallis (robust): A_valence ~ condition
print(kruskal.test(A_valence ~ condition, data = df_asym))
```

```

##
## Kruskal-Wallis rank sum test
##
## data: A_valence by condition
## Kruskal-Wallis chi-squared = 2.486, df = 2, p-value = 0.2885
cat("\nPairwise Wilcoxon (BH-adjusted):\n")

##
## Pairwise Wilcoxon (BH-adjusted):
print(pairwise.wilcox.test(
  df_asym$A_valence,
  df_asym$condition,
  p.adjust.method = "BH"
))

##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: df_asym$A_valence and df_asym$condition
##
##           Human-AI Human-Human
## Human-Human 0.50      -
## AI-AI       0.40      0.97
##
## P value adjustment method: BH

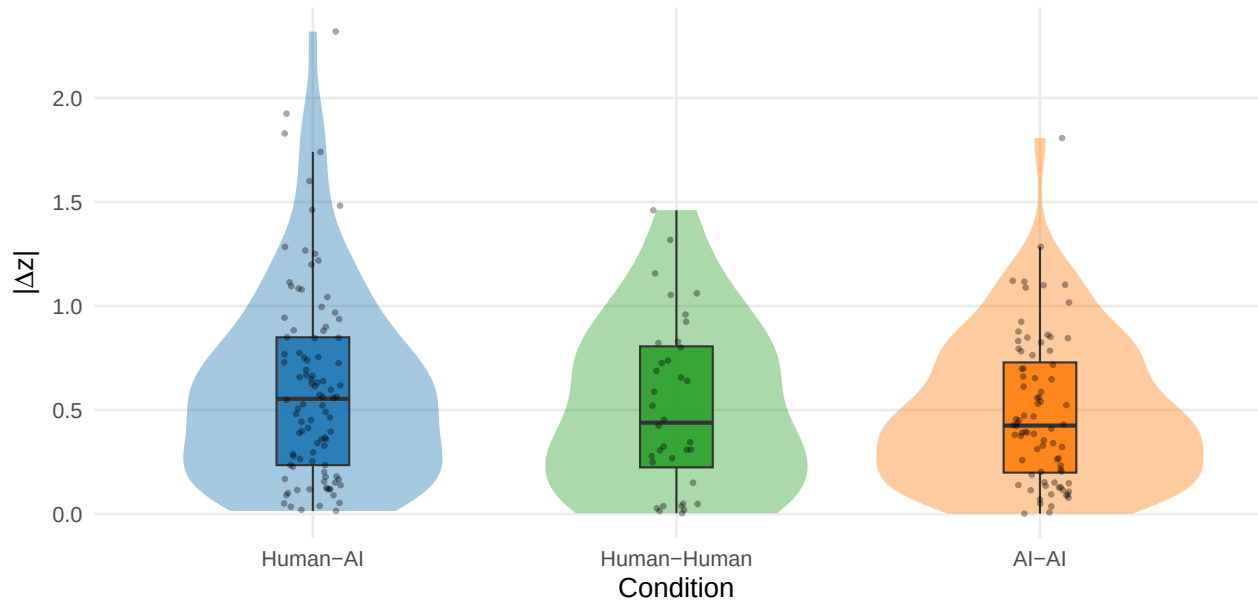
p_asym <- ggplot(df_asym, aes(x = condition, y = A_valence, fill = condition)) +
  geom_violin(alpha = 0.4, colour = NA) +
  geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.9) +
  geom_jitter(width = 0.08, alpha = 0.35, size = 0.9) +
  scale_fill_manual(values = condition_colors) +
  labs(
    title = "Per-story valence alignment asymmetry by condition",
    subtitle = "A_valence(i) = |atanh(r_first_to_second) - atanh(r_second_to_first)|",
    x = "Condition",
    y = expression("|" * Delta * z * "|")
  ) +
  theme(legend.position = "none")

print(p_asym)

```


Per-story valence alignment asymmetry by condition

$$A_valence(i) = |\text{atanh}(r_first_to_second) - \text{atanh}(r_second_to_first)|$$



```
ggsave(
  file.path(GRAPHES_DIR, "valence_asymmetry_per_story.png"),
  p_asym,
  width = 9,
  height = 5,
  dpi = 300
)
```

8 Bootstrap CIs and Effect Sizes on A_valence

```
boot_A <- bootstrap_condition_means(df_asym, "A_valence", R = 10000)
boot_A %>%
  knitr::kable(
    digits = 3,
    caption = "Bootstrap 95% CI on per-condition mean of A_valence (R = 10,000)"
  )
```

Table 13: Bootstrap 95% CI on per-condition mean of A_valence
(R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	98	0.612	0.523	0.708
Human-Human	36	0.516	0.389	0.647
AI-AI	80	0.490	0.416	0.569

```
effect_sizes_pairwise(df_asym, "A_valence") %>%
  knitr::kable(
    digits = 3,
    caption = "Pairwise Cliff's delta on A_valence"
  )
```

Table 14: Pairwise Cliff's delta on A_valence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	98	36	0.110	negligible
Human-AI vs AI-AI	98	80	0.132	negligible
Human-Human vs AI-AI	36	80	0.004	negligible

9 Planned Contrast: HA vs (HH + AA)/2

```
planned_contrast_HA_vs_same(df_asym, "A_valence") %>%
  knitr::kable(
    digits = 3,
    caption = "Planned contrast: A_valence in HA vs the average of HH and AA"
  )
```

Table 15: Planned contrast: A_valence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.109	0.062	1.769	156.883	0.079	-0.013	0.231

10 Directional Stability of Signed Valence Asymmetry

Tests whether the *direction* of the chronological first vs second alignment difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \text{atanh}(r_{\text{first} \rightarrow \text{second}, i}) - \text{atanh}(r_{\text{second} \rightarrow \text{first}, i})$. HA-specific identity-tied directionality predicts high **stability** and **sign_consistency** in HA, low in HH and AA, even when unsigned A_valence is similar.

```
ds_val <- directional_stability(df_asym, "signed_z_diff")
ds_val$by_condition %>%
  knitr::kable(
    digits = 3,
    caption = "Per-condition directional stability of signed valence asymmetry"
  )
```

Table 16: Per-condition directional stability of signed valence asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	98	0.192	0.751	0.256	0.582	0.013
Human-Human	36	-0.020	0.662	0.030	0.500	0.860
AI-AI	80	0.084	0.602	0.139	0.588	0.217

```
cat(sprintf(
  "Brown-Forsythe variance-equality p = %.4f\n", ds_val$variance_equality_p
))
```

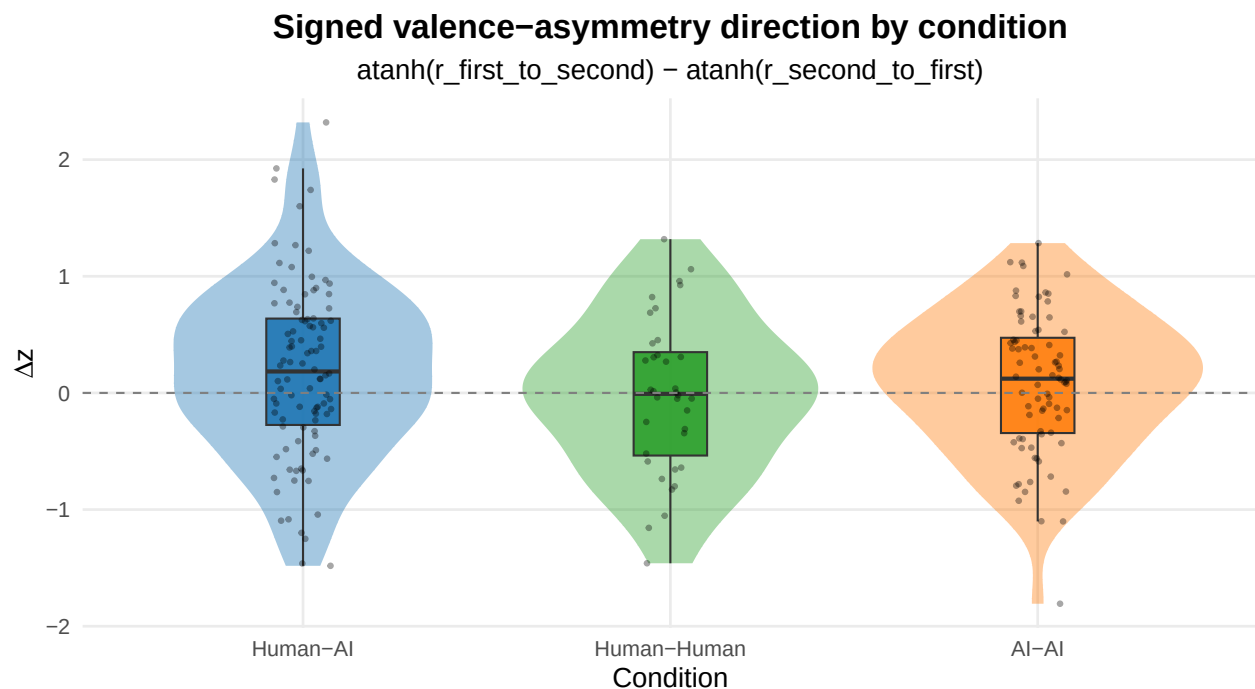
```
## Brown-Forsythe variance-equality p = 0.1587
```

```

p_signed <- ggplot(df_asym, aes(x = condition, y = signed_z_diff, fill = condition)) +
  geom_violin(alpha = 0.4, colour = NA) +
  geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.9) +
  geom_jitter(width = 0.08, alpha = 0.35, size = 0.9) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  scale_fill_manual(values = condition_colors) +
  labs(
    title = "Signed valence-asymmetry direction by condition",
    subtitle = "atanh(r_first_to_second) - atanh(r_second_to_first)",
    x = "Condition",
    y = expression(Delta * z)
  ) +
  theme(legend.position = "none")

print(p_signed)

```



```

ggsave(
  file.path(GRAPH_DIR, "valence_signed_asymmetry_direction.png"),
  p_signed,
  width = 9,
  height = 5,
  dpi = 300
)

```

11 Permutation Test on $A_{\text{valence}} \sim \text{condition}$

```

permutation_test_A(df_asym, "A_valence", R = 10000) %>%
  knitr::kable(
    digits = 4,
    caption = "Permutation test on  $A_{\text{valence}} \sim \text{condition}$  (R = 10,000 story-label permutations)"
  )

```

Table 17: Permutation test on $A_{\text{valence}} \sim \text{condition}$ ($R = 10,000$ story-label permutations)

F_obs	p_perm	R
2.003	0.1407	10000

12 Summary

Primary cross-condition result: per-story $A_{\text{valence}} = |\text{atanh}(r_{\text{first} \rightarrow \text{second}, i}) - \text{atanh}(r_{\text{second} \rightarrow \text{first}, i})|$, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of A_{valence}
- pairwise Cliff’s δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- 10k story-label permutation test on $A_{\text{valence}} \sim \text{condition}$

The Human-AI signed Human/AI directional decomposition lives in `analysis/human-ai/valence_alignment_analysis.Rmd` and is not duplicated here.

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-18 17:13:49.817592